Mathematics for Cryptography III: Linear Algebra

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Blockstream Research

research.blockstream.com

github.com/BlockstreamResearch/zk-book



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Motivation

Why Linear Algebra?

In cryptography we commonly need to work with a large set of *linear* relations, e.g. $\alpha_1 x_1 + \alpha_2 x_2 + \cdots + \alpha_n x_n = 0$ over a field $\mathbb{F}$.

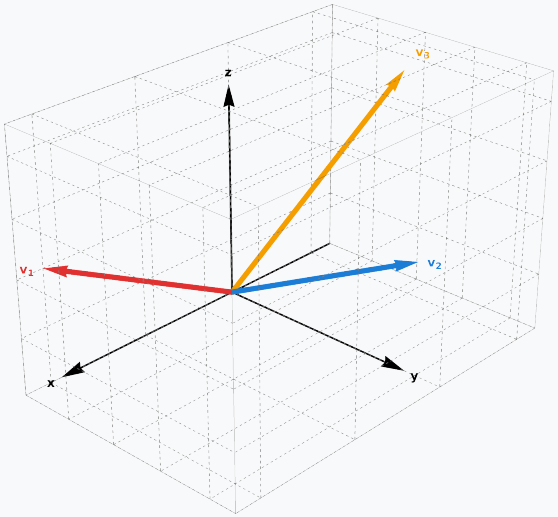
We collect variables into **vectors** (in bold): $\mathbf{x} = (x_1, \ldots, x_n)$, and relations into **matrices**:

$$\begin{cases} \alpha x_1 + \beta x_2 = 0 \\ \gamma x_1 + \delta x_3 = 0 \\ \varepsilon x_2 + \zeta x_3 = 0 \end{cases} \iff A\mathbf{x} = \mathbf{0}, \quad A = \begin{bmatrix} \alpha & \beta & 0 \\ \gamma & 0 & \delta \\ 0 & \varepsilon & \zeta \end{bmatrix}.$$

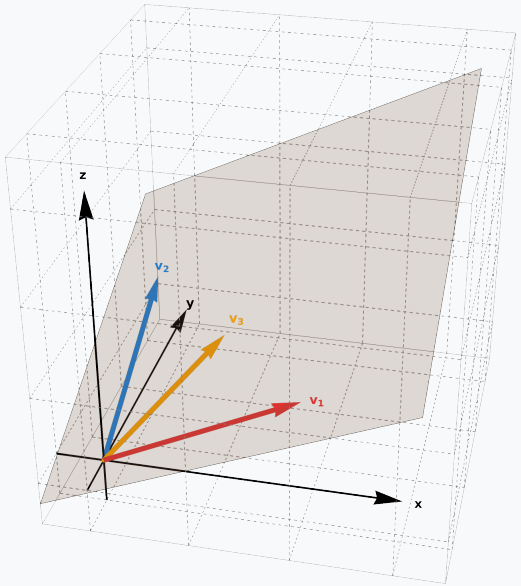
Where we use it

Linear PCPs and R1CS (Hadamard products of large vectors), inner-product arguments (Bulletproofs), and lattices (short solutions of $A\mathbf{x} = \mathbf{0}$).

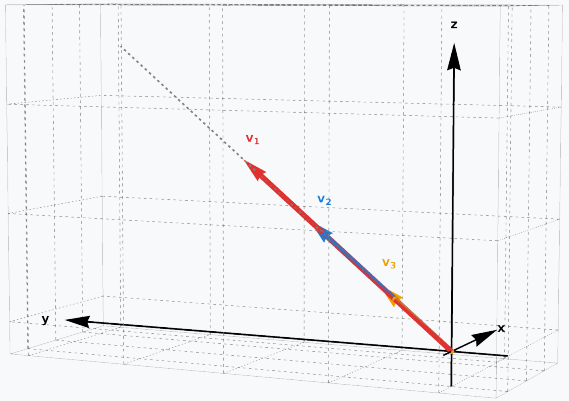
Geometric View



(a) vectors span $\mathbb{R}^3$: a solution always exists.



(b) vectors on a plane: infinitely many or no solutions.

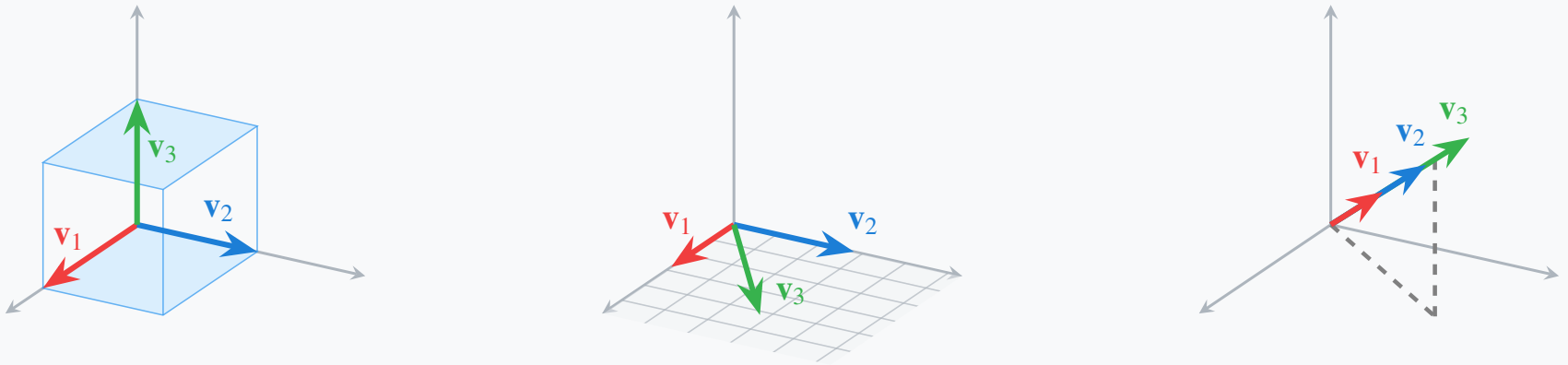


(c) vectors on a line: a solution exists only on this line.

Solutions of $\mathbf{v}_1 x_1 + \mathbf{v}_2 x_2 + \mathbf{v}_3 x_3 = \mathbf{t}$ depend on how $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3$ are related.

Determinant: the Volume Spanned by Columns

For $A = (\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3)$, $\det A$ is the (signed) volume of the parallelepiped spanned by the columns.



Takeaway

A plane or a line has no volume ($\det A = 0$): the equation $A\mathbf{x} = \mathbf{t}$ has either no or infinitely many solutions. If $\det A \neq 0$, there is always exactly one.

Vector Spaces

Vector Space

Definition

A **vector space** V over the field $\mathbb{F}$ is an abelian group for addition $+$ together with a scalar multiplication $\mathbb{F} \times V \to V$, $(\lambda, \mathbf{x}) \mapsto \lambda\mathbf{x}$, such that for any $\mathbf{v}, \mathbf{u} \in V$ and $\lambda, \mu \in \mathbb{F}$:

- $\lambda(\mathbf{u} + \mathbf{v}) = \lambda\mathbf{u} + \lambda\mathbf{v}$
- $(\lambda + \mu)\mathbf{v} = \lambda\mathbf{v} + \mu\mathbf{v}$
- $(\lambda\mu)\mathbf{v} = \lambda(\mu\mathbf{v})$
- $1 \cdot \mathbf{v} = \mathbf{v}$

Example

$\mathbb{F}^n$ with coordinate-wise operations is a vector space. So is $V_3 := \{\mathbf{x} \in \mathbb{F}^n : x_1 + x_2 + \cdots + x_n = 0\}$: it is closed under addition and scalar multiplication.

Remark

Polynomials $\mathbb{R}[T]^{\leq n}$ also form a vector space over $\mathbb{R}$.

Linear Subspace and Span

Definition

Let V be a vector space over $\mathbb{F}$ and $W \subseteq V$. W is a **linear subspace** of V if for any $\mathbf{w}_1, \mathbf{w}_2 \in W$ and $\alpha, \beta \in \mathbb{F}$ one has $\alpha\mathbf{w}_1 + \beta\mathbf{w}_2 \in W$.

Example

For vectors $\mathbf{v}_1, \ldots, \mathbf{v}_n \in V$, the **span**

$$\operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_n) := \{\lambda_1\mathbf{v}_1 + \cdots + \lambda_n\mathbf{v}_n : \lambda_1, \ldots, \lambda_n \in \mathbb{F}\}$$

is a linear subspace of V.

Question

Show that the set of solutions $\mathbf{x} \in \mathbb{F}^n$ of $\mathbf{u}_1 x_1 + \cdots + \mathbf{u}_n x_n = \mathbf{0}$ is a linear subspace of $\mathbb{F}^n$.

Subspaces: Examples

Example

Let $n \geq 2$ and

$$U := \{\mathbf{x} \in \mathbb{F}^n : x_1 = x_2 = 0\}, \quad W := \{\mathbf{x} \in \mathbb{F}^n : x_1 + x_2 = 0\}.$$

Both U and W are linear subspaces of $\mathbb{F}^n$, and U is a subspace of W.

Question

Let $V := \{\mathbf{z} \in \mathbb{F}^n : \mathbf{z} \odot (\mathbf{1}_n - \mathbf{z}) = \mathbf{0}\}$ (bit-vectors, see the Hadamard product below). When is V a linear subspace of $\mathbb{F}^n$?

Basis and Dimension

Linear Independence

Definition

Vectors $\mathbf{v}_1, \ldots, \mathbf{v}_n \in V$ are **linearly independent** if the only scalars $\lambda_1, \ldots, \lambda_n \in \mathbb{F}$ with

$$\lambda_1 \mathbf{v}_1 + \lambda_2 \mathbf{v}_2 + \cdots + \lambda_n \mathbf{v}_n = \mathbf{0}$$

are $\lambda_1 = \cdots = \lambda_n = 0$. Otherwise, they are **linearly dependent**.

Example

$\mathbf{v}_1 = (1, 0, 0)$, $\mathbf{v}_2 = (0, 1, 0)$, $\mathbf{v}_3 = (0, 0, 1)$ are linearly independent in $\mathbb{R}^3$.

Example

$\mathbf{u}_1 = (1, 2, 3)$ and $\mathbf{u}_2 = (2, 4, 6)$ are linearly dependent: $2\mathbf{u}_1 - \mathbf{u}_2 = \mathbf{0}$.

Basis and Dimension

Definition

Vectors $\mathbf{e}_1, \ldots, \mathbf{e}_n$ are a **basis** of V if $\text{span}(\mathbf{e}_1, \ldots, \mathbf{e}_n) = V$ and $\mathbf{e}_1, \ldots, \mathbf{e}_n$ are linearly independent.

Example

$\mathbf{e}_1 = (1, 0)$, $\mathbf{e}_2 = (0, 1)$ is the *canonical* basis of $\mathbb{F}^2$, but $\mathbf{u}_1 = (1, 2)$, $\mathbf{u}_2 = (2, 1)$ is a basis of $\mathbb{R}^2$ as well. For $\mathbb{F}[T]^{\leq n}$, the canonical basis is $1, T, \ldots, T^n$.

Lemma

Any two bases of a vector space have the same length.

Definition

The **dimension** $\dim V$ is the length of any basis of V. For example, $\dim \mathbb{F}^n = \dim \mathbb{F}[T]^{\leq n-1} = n$.

Dimension: Questions

Question #1

Let $V := \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$. Compute $\dim V$.

Question #2

Is $3,\ T + 2,\ T^2 + 4$ a basis of $\mathbb{F}_5[T]^{\leq 2}$?

Question #3

How many distinct 1-dimensional linear subspaces (lines through the origin) exist in $\mathbb{F}_3^3$?

Matrices

Matrix Operations

A matrix $A \in \mathbb{F}^{m \times n}$ has m rows and n columns; we write $A = (a_{i,j})$.

Definition (Matrix operations)

- **Addition** and **scalar multiplication** are element-wise.
- For $A \in \mathbb{F}^{m \times n}$ and $B \in \mathbb{F}^{n \times p}$, the **product** $C := AB \in \mathbb{F}^{m \times p}$ has elements $c_{i,j} := \sum_{\ell=1}^{n} a_{i,\ell} b_{\ell,j}$.
- The **transpose** $A^\top \in \mathbb{F}^{n \times m}$ has elements $A^\top_{i,j} = A_{j,i}$.

Example

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}, \; B = \begin{bmatrix} 2 & 1 \\ 1 & 3 \\ 1 & 1 \end{bmatrix} \implies AB = \begin{bmatrix} 5 & 6 \\ 7 & 9 \end{bmatrix}, \; BA = \begin{bmatrix} 4 & 4 & 5 \\ 7 & 7 & 5 \\ 3 & 3 & 3 \end{bmatrix}.$$

Matrix-Vector Product

Vectors are written in columns. For $A \in \mathbb{F}^{m \times n}$ and $\mathbf{x} \in \mathbb{F}^n$:

$$A\mathbf{x} = \begin{bmatrix} a_{1,1} & \dots & a_{1,n} \\ \vdots & \ddots & \vdots \\ a_{m,1} & \dots & a_{m,n} \end{bmatrix} \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} a_{1,1}x_1 + \cdots + a_{1,n}x_n \\ \vdots \\ a_{m,1}x_1 + \cdots + a_{m,n}x_n \end{bmatrix}.$$

Remark

Why is the matrix product so odd? We study maps $\mathbf{x} \mapsto A\mathbf{x}$. Composing $\mathbf{x} \mapsto B\mathbf{x}$ and then $\mathbf{y} \mapsto A\mathbf{y}$ must again be such a map, $\mathbf{x} \mapsto C\mathbf{x}$, and $C = AB$ is the only way to get it.

Question

Let $\mathbf{x} = (1, 2, 1)$. Compute $\mathbf{x}^\top \mathbf{x}$ and $\mathbf{x}\mathbf{x}^\top$. For $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$, compute A^n.

Linear Systems and Rank

A system of linear equations is just $A\mathbf{x} = \mathbf{t}$:

$$\begin{cases} x_1 + 2x_2 + 3x_3 = 2 \\ 2x_1 + x_2 + 3x_3 = 1 \\ -x_1 + x_2 + x_3 = 0 \end{cases} \iff \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ -1 & 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}.$$

Definition

The **rank** of $A = (\mathbf{v}_1, \ldots, \mathbf{v}_n)$ is $\operatorname{rank}(A) := \dim \operatorname{span}(\mathbf{v}_1, \ldots, \mathbf{v}_n)$. Row rank and column rank are always equal.

Proposition

For $A \in \mathbb{R}^{n \times n}$ and $\mathbf{t} \in \mathbb{R}^n$:

1. $A\mathbf{x} = \mathbf{t}$ always has a single solution iff $\det A \neq 0$;
2. *(Rouché-Capelli)* a solution exists iff $\operatorname{rank}(A \,||\, \mathbf{t}) = \operatorname{rank}(A)$.

Inner Product

Inner Product Spaces

Definition

An **inner product** on V is a function $\langle \cdot, \cdot \rangle : V \times V \to \mathbb{F}$ that is **positive** ($\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$), **definite** ($\langle \mathbf{v}, \mathbf{v} \rangle = 0$ iff $\mathbf{v} = \mathbf{0}$), **additive**, **homogeneous** and **symmetric**.

Example

The **dot product** on $\mathbb{F}^n$: $\langle \mathbf{u}, \mathbf{v} \rangle := \mathbf{u}^\top \mathbf{v} = \sum_{i=1}^{n} u_i v_i$. For $\mathbf{u} = (1, 2, 3)$, $\mathbf{v} = (2, 4, 3)$, we get $\langle \mathbf{u}, \mathbf{v} \rangle = 2 + 8 + 9 = 19$.

Definition

The **norm** of $\mathbf{v}$ is $\|\mathbf{v}\| := \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$. Vectors $\mathbf{v}, \mathbf{u}$ are **orthogonal** if $\langle \mathbf{v}, \mathbf{u} \rangle = 0$.

Remark

Over finite fields or $\mathbb{Z}_N$, "≥ 0" is meaningless, so we only require additivity, homogeneity and symmetry.

Inner Products: More Examples

Example

A symmetric matrix $Q \in \mathbb{R}^{n \times n}$ is *positive-definite* if $\mathbf{x}^\top Q\mathbf{x} > 0$ for any non-zero $\mathbf{x}$. Then $\langle \mathbf{x}, \mathbf{y} \rangle_Q := \mathbf{x}^\top Q\mathbf{y}$ is an inner product.

Example

Polynomials have "lengths" too: on $V := \mathbb{R}[T]^{\leq 2}$, define $\langle p, q \rangle := \int_0^1 p(\tau)q(\tau)\, d\tau$.

Question

Compute the norm $||T||$ in V with the inner product above. Find any $q \in V$ orthogonal to T^2.

Hadamard and Outer Products

Hadamard Product

Definition

For $A, B \in \mathbb{F}^{m \times n}$, the **Hadamard product** $A \odot B$ is the matrix C with $C_{i,j} = A_{i,j} B_{i,j}$: we multiply element-wise.

Example

$$\begin{bmatrix} 1 & 1 & 2 \\ 3 & 0 & 3 \end{bmatrix} \odot \begin{bmatrix} 3 & 2 & 1 \\ 0 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 3 & 2 & 2 \\ 0 & 0 & 3 \end{bmatrix}.$$

Why it matters

To say that $\mathbf{z} \in \mathbb{F}^n$ is a bit-vector (every $z_i \in \{0, 1\}$), write $\mathbf{z} \odot (\mathbf{1}_n - \mathbf{z}) = \mathbf{0}$. R1CS is also a Hadamard-product relation.

Outer Product

Definition

For $\mathbf{u} \in \mathbb{F}^n$, $\mathbf{v} \in \mathbb{F}^m$, the **outer product** is $\mathbf{u} \otimes \mathbf{v} := \mathbf{u}\mathbf{v}^\top \in \mathbb{F}^{n \times m}$, with entries $u_i v_j$.

Example

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \otimes \begin{bmatrix} 2 \\ 4 \\ 3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} 2 & 4 & 3 \end{bmatrix} = \begin{bmatrix} 2 & 4 & 3 \\ 4 & 8 & 6 \\ 6 & 12 & 9 \end{bmatrix}.$$

Lemma

The outer product is distributive, commutes with scalars, $\mathbf{u} \otimes \mathbf{v} = (\mathbf{v} \otimes \mathbf{u})^\top$, and for non-zero $\mathbf{u}, \mathbf{v}$ it is a rank-1 matrix.

Thank you for your attention



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